Hu et al. 2026 — ELF: Embedded Language Flows

Continuous flow matching in embedding space; weight-tied unembedding decoder | arXiv:2605.10938 | my study repo

One-Liner

Run continuous-time flow matching in a frozen pretrained embedding space; reuse the same network's final timestep as the decoder via the tied unembedding matrix. Minimal-machinery DLM that beats discrete (MDLM, SEDD) and prior continuous (Diffusion-LM) baselines at $\sim 10\times$ fewer training tokens and fewer sampling steps.

What's Novel

- **No separate decoder.** Prior continuous DLMs train one; ELF observes the flow's final step IS the decoder if the unembedding W_E is weight-tied to the input embedding.
- **Stays continuous until $t = 1$.** Most continuous DLMs supervise discretization at every step (per-step cross-entropy). ELF lets the flow run free in $\mathbb{R}^d$ and only discretizes at the final step.
- **CFG plugged in cleanly.** Structural symmetry with image-domain flow matching means classifier-free guidance transfers directly. ELF uses training-time CFG to avoid the $2\times$ inference overhead.
- **Pretrained contextual embeddings dominate.** Ablation: pretrained > non-contextual > frozen-random. The embedding manifold is doing real work.

Math in 60 Seconds

Linear interpolation path with $z_1 = E(x)$ the clean embedding and $z_0 \sim \mathcal{N}(0, I)$:

$$z_t = (1-t)z_0 + tz_1, \quad t \in [0,1].$$

Flow-matching training objective:

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{x,z_0,t}\big[\|v_\theta(z_t,t) - (z_1 - z_0)\|^2\big].$$

Final-step discretization via weight-tied unembedding W_E:

$$\pi_\theta(z_{t\to 1}) = \mathrm{softmax}(W_E\, \hat{x}_\theta(z_{t\to 1}, 1)).$$

CFG combines conditional and unconditional flows with a single guidance scalar ω:

$$v_{\mathrm{cfg}}(z_t \mid c) = \omega\, v_\theta(z_t \mid c) + (1-\omega)\, v_\theta(z_t \mid \emptyset).$$

What I'd Push Back On

- Pretrained embeddings carry a lot of the win; the paper doesn't decompose embedding-vs-flow contributions. A discrete-DLM baseline given the same pretrained embedding space would settle this.
- Generative perplexity under GPT-2 Large biases the metric toward GPT-2-like outputs. Unigram entropy as the diversity proxy doesn't catch n-gram repetition.
- No matched-compute autoregressive baseline. The interesting question — does ELF beat a 105M-param decoder-only transformer at the same training budget? — is unanswered.
- "Minimal adaptation" undersells self-conditioning, training-time CFG, and the embedding choice. Minimal in form, several engineering pieces in practice.

My One Proposed Extension

Scheduled CFG: replace constant ω with a time-varying $\omega(t)$ that emphasises guidance near the boundaries of the path. Concretely, $\omega(t) = \omega_{\max} \cdot \big(2t(1-t) + \epsilon\big)^{-\beta}$ (U-shape, peaks at $t \to 0$ and $t \to 1$). The intuition: pure noise at the start and discrete-decoding at the end are the high-stakes regions; the middle of the path is where the flow is most reliable on its own. A prototype on a toy 2D Gaussian mixture (`improvements/cfg-schedule.py`) shows U-shape can improve both quality and diversity vs fixed-ω at the same sampling cost. Inference-only modification — no retraining required to test on existing ELF checkpoints. Testable with a CFG-scale sweep on OpenWebText.

Full proposal set: `05-improvements.tex` in the study repo.

My One Question

If ELF's design philosophy is "operate in continuous embedding space; discretize only at the final step," what's the right notion of "intermediate trajectory quality"? Is the flow passing through linguistically meaningful intermediate states, or just along a Euclidean geodesic in an embedding manifold with semantic structure orthogonal to the path?